Reconstructing and Analysing a forgotten World: A 3D-Positioning and Annotation System for the Paintings of the Kucha Project

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Abstract

The Buddhist cave complexes of Kucha in Xinjiang preserve extensive mural programmes dating from the fifth to the tenth centuries. During early twentieth-century expeditions, large parts of these murals were removed, resulting in the worldwide dispersal of fragments and the loss of their original spatial context (Yaldiz, 1987; Popova, 2008; Dreyer, 2015; Zhao, 2021). Reconstructing these contexts remains a major challenge.

The project *Buddhist Murals of Kucha on the Northern Silk Road* addresses this challenge through a database-supported information system that links in-situ paintings, dispersed fragments, historical photographs, and excavation reports. One of the central research questions is how iconographic information can be integrated with spatial context to support the reconstruction and analysis of complex mural programmes.

The project employs the Annotorious[1] annotation framework together with a controlled vocabulary – the Kucha Murals Thesaurus – comprising approximately 1,200 iconographic

[1] https://annotorious.dev/

categories. At present, the system contains nearly 11,000 annotated elements.[2]

The KMT serves as the semantic backbone of the image annotation process. It is driven directly by the visual evidence preserved in the murals and is continuously curated and maintained by project team. The vocabulary is systematically structured into three main top-level categories: iconography, ornaments, and pictorial elements.(Radisch, 2026)

To ensure broad interoperability and adherence to FAIR principles, we are currently in the process of cross-referencing KMT entries with established external authority files such as Wikidata, the Getty Art & Architecture Thesaurus (AAT), and the vocabularies of the DiGA (Digitization of Gandharan Artifacts) project. Building upon these initial linkages, our current development phase focuses on transitioning the KMT into a formal, fully compliant SKOS ontology, following the structural model established by the DiGA project. (Radisch/Fellner, 2025)

Since the original murals were organised within square and rhombic register systems, these structures provide the spatial framework for the reconstruction process (Figure 1). Within this coordinate system, horizontal rows along the y-axis are termed "registers", while vertical columns along the x-axis are termed "numbers". Previously recorded as static textual metadata, this grid system is first transformed into an interactive visualisation where a customizable number of rectangular or rhombic fields are projected onto a blank square canvas.

To integrate these registers into the 3D reconstruction workflow, this 2D canvas is subsequently mapped onto a specific 3D mesh of a cave model. Because the architectural wall meshes are rarely perfectly rectangular, any areas of the square canvas that cannot be projected onto the mesh are automatically blacked out upon a successful connection. This visual feedback mechanism enables researchers to precisely align the abstract register systems with the irregular structures of the physical walls. The resulting schematic 3D cave provides precise information about the spatial position of individual paintings and their relationships to other objects. Furthermore, the system is fully interactive,

[2] https://kucha.saw-leipzig.de/

allowing users to open pop-ups with detailed information for each individual register entry. (Radisch, 2026)

Preview images derived from annotations encompassing complete paintings are displayed within their corresponding register positions, providing immediate visual orientation. In a subsequent step, SVG-based annotations are also mapped directly onto the preview images. A hierarchical tree of iconographic categories is integrated into the 3D environment; selecting a category highlights all associated motifs, enabling exploratory analysis of the pictorial programme. When an annotation belongs to a painting that is not currently displayed within the selected register position, the entire register area is highlighted instead. (Figure 2)

The resulting system supports both visual contextualisation and content-based spatial analysis. Researchers can investigate relationships between motifs, trace compositional structures, and compare iconographic distributions across walls, registers, and multiple caves. While interactive 3D heritage platforms such as Kompakkt support the annotation of three-dimensional objects through point-based references (Eide et al., 2019), our approach enables surface-specific visualisation of polygon-based annotations directly within the reconstructed space. Likewise, projects such as Florence As It Was successfully reconstruct artworks within their original architectural settings (Bent et al., 2022), but do not provide direct access to detailed pictorial annotations within those visualisations. By projecting semantic polygon annotations onto wall surfaces, individual figures, objects, and narrative elements can be explored in relation to their architectural context rather than as isolated images.

At present, however, the implementation remains closely tied to the underlying annotation and visualisation framework, limiting straightforward reuse in other projects. This limitation stems primarily from the current lack of a universally accepted standard for feeding surface annotation data into 3D viewers. Addressing this desideratum represents a significant opportunity for the FAIR 3D Heritage community. The development of a unified standard—for example, as part of emerging IIIF 3D specifications or through the further development of existing standards and formats — would be instrumental in enabling existing 3D viewers to natively support complex surface annotations.

However, this prospect immediately raises the question of how such surface annotations should be stored, exposing a fundamental semantic and architectural dilemma. On the one hand, established standards bind annotations directly to a 2D source image. Integrating this image-centric approach into a future 3D standard would strongly support interoperability, yet rendering dynamic SVG points within a 3D environment is computationally expensive. On the other hand, storing translated SVG coordinates as UV map coordinates allows them to be passed directly to a shader, which guarantees fast processing and highly efficient rendering. Yet, this performance comes at the cost of interoperability: the annotation points are decoupled from the coordinate space of the underlying source image and become bound to the specific 3D mesh. By presenting this dilemma, this contribution aims to initiate a collaborative discussion within the community, working jointly towards a sustainable, community-centred approach.

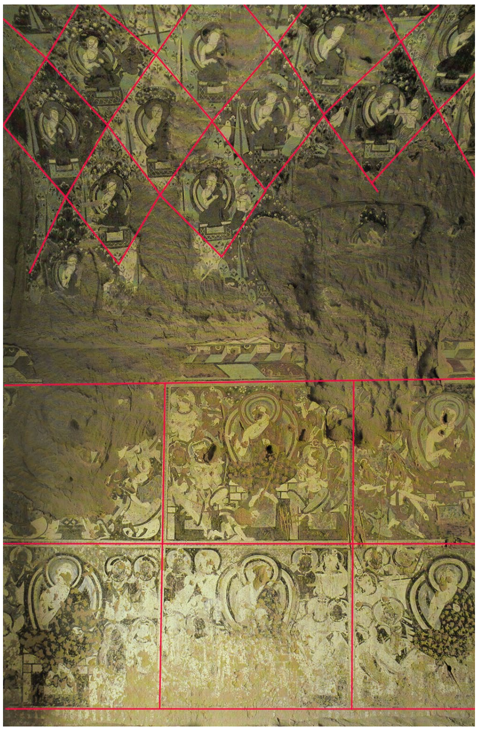

Figure1: Example of the register systems: above in rhombic form, below in square form. Source: Kumtula Grottoes, 1985.

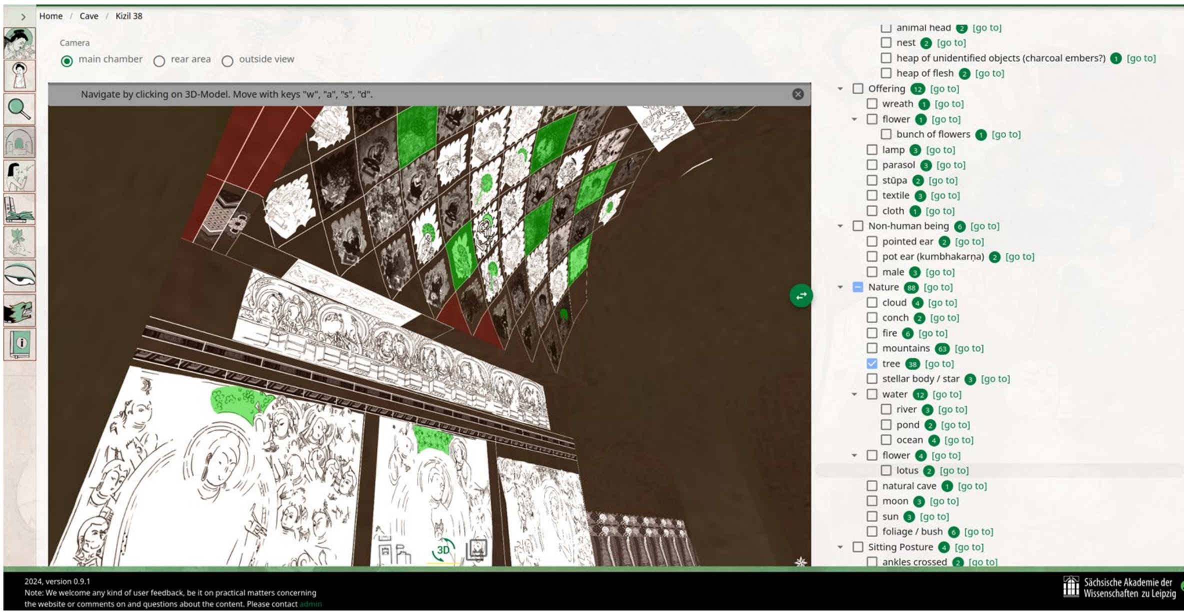

Figure2: Example of integrated Polygon annotation in 3D-View. If a polygon annotation is on another picture, the hole area is highlighted instead Source: https://kucha.saw-leipzig.de/cave/kizil%2038

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